

ParaTrace: Large language model rewriting induces systematic loss of dementia-related linguistic biomarkers in clinical speech

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Over 772,000 Canadians are living with dementia, while early detection remains a central objective of Canada's National Dementia Strategy. Computational analysis of spontaneous speech is promising because cognitive decline can manifest not only in what a patient communicates, but also in how it is expressed. In Ontario, ambient AI documentation is already entering clinical workflows: The Ottawa Hospital has adopted Microsoft Dragon Copilot following an ambient-AI pilot, while Ontario's AI Scribe Program maintains a Vendor of Record with 29 prequalified suppliers. This raises an information-fidelity question beyond privacy and consent. ParaTrace evaluates whether large language model rewriting preserves cognitive-linguistic features used in cognitive-status classification. Using 552 clinically labelled DementiaBank Pitt transcripts from 292 participants (243 control transcripts; 309 dementia transcripts), we extracted 20 features spanning lexical diversity, repetition, semantic coherence, syntactic complexity, idea density, word-finding, vocabulary sophistication, and content-unit measures. Each transcript was evaluated at five levels: L0 (original), L1 (grammar correction), L2 (light paraphrase), L3 (moderate rewrite), and L4 (full reformulation). L1–L4 were generated independently using OpenAI GPT-4o-mini and Anthropic Claude Sonnet 4.6, producing 4,416 rewritten samples. Participant-grouped stratified five-fold cross-validation used a Random Forest trained on L0 features and evaluated on aligned held-out participants. Accuracy declined from 71.7% at L0 to 65.0%, 57.0%, 49.9%, and 48.4% at L1–L4; balanced accuracy fell to 47.9% at L4. At L2, 19 of 20 features were significantly altered for both backends after paired Wilcoxon signed-rank testing with Benjamini-Hochberg correction, despite semantic cosine similarity of 0.864 for Anthropic and 0.849 for OpenAI. These findings show that high semantic similarity can coexist with degradation of predictive linguistic structure. We therefore propose extracting clinically relevant linguistic representations from raw speech before generative rewriting and retaining them alongside the polished note.